Jinil Kim

haba6030@snu.ac.kr | Homepage | GitHub | LinkedIn

RESEARCH INTERESTS

- *Computational Neuroscience* (computational principles and latent structure shared by brain and behavior)
- *Neuro-Inspired AI and AI for Neuroscience* (multimodal predictive models)
- *Computational Psychiatry* (model-derived markers; predicting intervention response)

EDUCATION

Seoul National University Seoul, South Korea
B.S. Computer Science · B.A. Psychology · B.A. Linguistics Mar 2021 – Exp. Feb 2027
GPA: 4.18 / 4.3 (3.97 / 4.0) · 9 semesters (Mandatory military service, Jul 2023 – Jan 2025)

The University of Hong Kong Hong Kong SAR
Summer Research Program, Faculty of Psychology Jun 2025 – Aug 2025

PUBLICATIONS & MANUSCRIPTS

Workshop Papers

- **Kim, J.**, & Ryu, J. (2026). *Conversational relevance in the conjunction fallacy is represented independently in language models.* Accepted for Conference on Neural Information Processing Systems (NeurIPS) 2026 Workshop on Linguistic Principles for Foundation Models, Sydney, Australia. [paper]
- **Kim, J.**, Cho, A. M., Seo, J., & Cha, J. (2026). *Inferring individualized color-vision distortions from fMRI hue-representation geometry.* Presented at International Conference on Machine Learning (ICML) 2026 Workshop on Structured Data for Health, Seoul, Korea. [paper]

Under Review

- **Kim, J.**, Jeong, J., Yun, H., Oh, M.-H., & Ahn, W.-Y. (2026). *Inverse reinforcement learning with state-dependent foresight for human decision making.* Under review at the International Conference on Learning Representations (ICLR).

In Preparation

- **Kim, J.**, Cho, A. M., Seo, J., & Cha, J. *Preserved categorical identification with impaired hue interpolation in color vision deficiency motivates a cortically derived correction filter.*
- **Kim, J.**, & Yao, Y.-W. *Depression relates to when, and not only how strongly, information enters a social decision.*

CONFERENCE PRESENTATION

- **Kim, J.**, Cho, M., Seo, J., & Cha, J. (2026). *fMRI decoding reveals intact neural color representations in color vision deficiency.* OHBM 2026 Annual Meeting, Bordeaux, France. [Poster]

RESEARCH EXPERIENCES

Connectome Lab Seoul National University
Undergraduate RA (Advisor: Prof. Jiook Cha) Apr 2025 – Present

- Led a project characterizing distortions of cortical hue geometry in color vision deficiency from fMRI. Translated them into personalized filters that moved hue-discrimination thresholds to the control range.
- Proposed an uncertainty quantification pipeline for an EEG neural field foundation model, measuring whether predicted uncertainty matches actual error and using it to guide electrode placement.

Computational Clinical Science Lab Seoul National University
Undergraduate RA (Advisor: Prof. Woo-Young Ahn) Apr 2025 – Present

- Developed SC-AIRL, an inverse RL framework with state-dependent foresight, raising action accuracy by up to 10 points over standard AIRL. Inferred planning depth correlated with anxiety and impulsivity.

MIND Lab University of Hong Kong
Visiting Researcher (Advisor: Prof. Yuan-Wei Yao) Jun 2025 – Aug 2025

- Adapted a time-varying drift-diffusion model to social decision data with an onset-latency parameter. Showed that the depressed-range group integrates the reward–emotion contrast 0.29 s later ($d = 0.32$).

SCHOLARSHIPS & HONORS

KOSAF Humanity 100 Years Undergraduate Scholarship ($12,500) Mar 2025 – Jul 2026
SNU Undergraduate Research Grant ($3,500) Apr 2025
Best Poster Award, SNU Undergraduate Research Program Jan 2026
SNU Student-Proposed Liberal Arts Course Contest (1st Place) Jul 2023

RELEVANT COURSEWORK

Neuroscience	Neuroscience · Data Science in Human Neuroimaging · Clinical Neuropsychology · Cognitive Processes · Computational Modeling in Experimental Psychology · Brain, Mind, and Behavior
Computer Science	Data Structures · Algorithms · Computer Architecture · Logic Design · Computer Programming · Computational Linguistics
Mathematics & Statistics	Linear Algebra · Discrete Mathematics · Calculus I–II · Psychological Statistics · Advanced Psychological Statistics

ACADEMIC SERVICES

Conference Reviewer: NeurIPS 2026 Workshop on Foundation Models for the Brain and Body
Volunteer: ICML 2026

TEACHING EXPERIENCE

Basic Computing: First Adventure in Computing Seoul National University
Teaching Assistant (Instructors: Yamada Akihiko, Heajung Min) Autumn 2025 · Spring 2026
Instructed laboratory classes regarding Python and basic programming for local and international students; English-medium section in Spring 2026.

LANGUAGES

Programming: Python (advanced), R, C++, Java (proficient), Stan (intermediate)